



IDX G9 COMPUTER SCIENCE H STUDY GUIDE

ISSUE S2 FINALS

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AI (artificial intelligence)

- the technology that enables computers and machines to simulate human intelligence and problem-solving capabilities

- can be modeled after the decision-making processes of the human
- can 'learn' from available data and make increasingly more accurate classifications or predictions over time

- types

○ strong AI

- aka general AI, superintelligence
- a theoretical form of AI, where it has the same intelligence as humans and will surpass the ability of human brain
- possess self-awareness and consciousness
- able to solve problems, learn, and plan for the future

○ weak AI

- aka narrow AI

- AI trained to perform specific tasks
- Examples: Apple's Siri, self-driving vehicles

Machine learning

- an input layer, one or two 'hidden' layers, and an output layer
 - needs more feature engineering by human experts to help with feature extraction
- uses supervised learning: the data needs to be structured or labeled to enable the algorithm to extract features from the data
- Uses neural networks with a lot of layers to 'learn' from huge amounts of data
 - These neural networks are programmatic structures modeled after the decision-making processes of the human brain
 - Contains layers of interconnected of interconnected nodes to extract features from data and make predictions
- **Deep learning**
 - an input layer, three or more (but usually hundreds) of hidden layers, and an output layer
 - cost much more data than before, but labeled datasets are limited
 - unsupervised learning: they automate the extraction of features from large, unlabeled and unstructured data sets for saving human intervention
- 3 types
 - **Supervised Learning**
 - labels datasets to train models to classify data or predict outcomes as intended-
 - **K-nearest-neighbor (KNN) Algorithm**

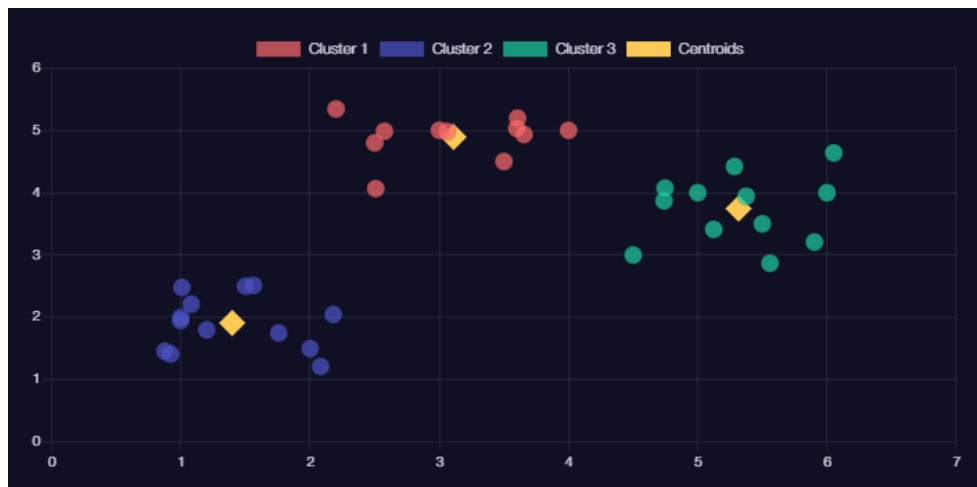
- Find the k nearest neighbors to determine which category the point belongs to

○ Unsupervised Learning

- learns from the input data itself, identifying patterns and structures in the data without the explicit labels or outcomes

▪ K-Means Clustering Algorithm

- repeatedly assigning points to clusters and updating those clusters' centers, till it can reach the center of clusters



○ Reinforcement Learning

- learns to perform a task by trial and error

- data types

○ Classification

- Discrete categories

○ Regression

- Continuous values

Determine Distance between query point and other points

- x & y are two points and xi & yi are the coordinate in each axis

- Euclidean Distance

- it measures a straight line between the query point and the other point being measured

$$d(x, y) = \sqrt{\sum_{i=1}^n (y_i - x_i)^2}$$

○

- Manhattan Distance

- measures the absolute value between two points

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

○

- Minkowski Distance

- The generalized form of Euclidean and Manhattan distance metrics

$$d(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

○

- Hamming Distance

- identifying the points where the vectors do not match

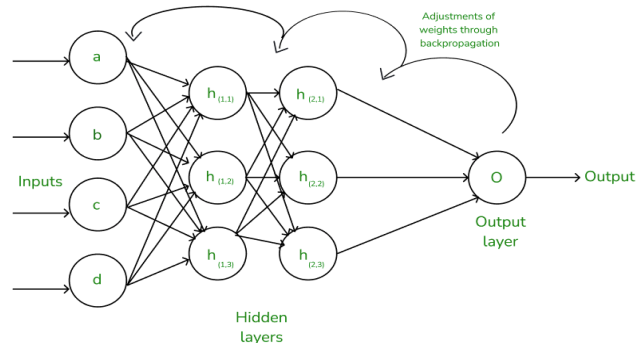
$$\text{Hamming Distance}(A, B) = \sum_{i=1}^n [A_i \neq B_i]$$

○

History of AI

- 1950: Alan Turing asks the following question: "Can machines think?"

- he offers a test, now famously known as the "Turing Test", where a human interrogator would try to distinguish between a computer and human text response
- 1956: John McCarthy coined the term "artificial intelligence"
 - at the first-ever AI conference at Dartmouth College
- 1967: Frank Rosenblatt built the Mark 1 Perceptron
 - the first computer based on a neural network that “learned” through trial and error
- 1980s: Neural networks which use a back-propagation algorithm to train itself become widely used in AI applications.



- 1995: Stuart Russell and Peter Norvig published The First Textbook about AI
 - Artificial Intelligence: A Modern Approach
 - delved into four potential goals or definitions of AI, which differentiates computer systems on the basis of rationality and thinking vs. Acting
- 1997: IBM's Deep Blue defeats chess champion Garry Kasparov
- 2016: DeepMind's AlphaGo beats Lee Sedol in Go
- 2023: Rise of large language models (LLMs) like ChatGPT.

Neural Network

- **Definition:** A type of machine learning modeled after the human brain, using interconnected nodes to learn through trial and error.

- **Structure:** Nodes (basic processing units) form layers that process different levels of detail in data.

Example 1: College Acceptance Model

- **Inputs:** Grades, class activities.
- **Hidden Layers:**
 - First layer focuses on grades (e.g., English/Math grades).
 - Second layer focuses on activities (e.g., club/sports activities).
- **Nodes:** Each node processes a specific aspect within its layer, with dense connections to all nodes in the next layer.

Example 2: Temperature Conversion Model

- **Simple Case:** Linear conversion from °C to °F with one input (°C) and one output (°F).
- **Formula:** Output = Weight × Input + Bias (e.g., initial incorrect weights: $F = 5.67 \times C + 12.5$; refined weights: $F = 1.828 \times C + 28.54$).
- **Complexity Warning:** Using more nodes (e.g., cubic equations) can overcomplicate simple linear problems.

Model Fitting Parameters

- **Number of Nodes:** Increases model complexity.
- **Number of Layers:** Enables deeper feature extraction.
- **Training Epochs:** Multiple passes to refine model accuracy.

Convolution Operation

- **Definition:** A mathematical operation combining two functions (e.g., input data and a kernel) to generate an output, widely used in signal processing and computer vision.
- **Example:** Kernel (e.g., 3×3 matrix) slides over input data, performing element-wise multiplication and summation to produce a new feature matrix.

Input: g					Kernel: f			Convolution: f * g					
3	5	2	8	1	1	0	0	3x1	5x0	2x0	8	1	
9	7	5	4	3	1	1	0	9x1	7x1	5x0	4	3	
2	0	6	1	6	0	0	1	2x0	0x0	6x1	1	6	
6	3	7	9	2				6	3	7	25	18	17
1	4	9	5	1				1	4	9	18	22	14
											20	15	23

Convolutional Neural Networks (CNNs)

- **Definition:** Feed-forward neural networks that learn feature engineering via filter (kernel) optimization.
- **Key Layers:**
 - **Convolution Layer:** Applies kernels to extract features.
 - **Subsampling/Pooling Layer:** Reduces spatial dimensions (e.g., maxpool).
 - **Fully Connected Layer:** Connects features for classification.
- **Feature Maps:** Outputs of convolution layers, representing learned features.

CNN in Computer Vision

- **Image Representation:**

- Digital photos as pixel arrays (RGB values 0–255).
- E.g., Digit "8" as a binary pixel array (0s and 1s).

- **Convolution Applications:**

- **Blurring:** Kernel replaces pixels with the mean of neighbors.
- **Edge Detection:** Kernel detects gradient changes (e.g., vertical edges).

- **Cat Classification Pipeline:**

Input $\rightarrow$ [Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool] $\rightarrow$ [Conv $\rightarrow$ ReLU $\rightarrow$ MaxPool] $\rightarrow$ Fully

Connected $\rightarrow$ Output"